Evolution

Evolutionary biology is the study of the history of life forms on earth, and to understand the changes in flora and fauna over millions of years we must have an understanding of the context of the origin of life.

The longest story

- that context is the evolution of earth, of stars and indeed of the universe itself
- what follows is the longest of all the construed and conjectured stories
- it is the story of the origin of life and of the evolution of life forms, that is biodiversity, on planet earth, against the background of the evolution of the universe

Looking back in time

- when we look at stars on a clear night sky we are, in a way, looking back in time, since stellar distances are measured in light years
- what we see today is an object whose emitted light started its journey millions of years back and from trillions of kilometres away, reaching our eyes now
- when we see objects in our immediate surroundings we see them instantly and hence in the present time, so when we see stars we apparently are peeping into the past

A unique event

- the origin of life is considered a unique event in the history of the universe
- the universe is vast, and relatively speaking the earth itself is almost only a speck

In this chapter

6.1 Origin of life
6.4 Evolution of life forms
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6.1 How the earth came to be

The Big Bang, and the early earth

- the universe is very old, almost 13.8 billion years old, huge clusters of galaxies comprise it, and galaxies contain stars and clouds of gas and dust
- the Big Bang theory attempts to explain the origin of the universe, and talks of a singular huge explosion unimaginable in physical terms, after which the universe expanded and hence the temperature came down, hydrogen and helium forming sometime later, the gases condensing under gravitation and forming the galaxies of the present day universe
- in the solar system of the milky way galaxy, earth was supposed to have been formed about 4.5 billion years back, and there was no atmosphere on early earth, water vapour, methane, carbondioxide and ammonia released from molten mass covering the surface
- the UV rays from the sun broke up water into hydrogen and oxygen, the lighter hydrogen escaped, oxygen combined with ammonia and methane to form water, CO_2 and others, the ozone layer was formed, and as it cooled the water vapour fell as rain to fill all the depressions and form oceans, life appearing 500 million years after the formation of earth, that is almost four billion years back

6.2 Where life came from

Panspermia and spontaneous generation

- some scientists believe life came from outside, early Greek thinkers having thought that units of life called spores were transferred to different planets including earth, and panspermia is still a favourite idea for some astronomers
- for a long time it was also believed that life came out of decaying and rotting matter such as straw and mud, which was the theory of spontaneous generation
- Louis Pasteur by careful experimentation demonstrated that life comes only from pre-existing life, showing that in pre-sterilised flasks life did not come from killed yeast, while in another flask open to air new living organisms arose from killed yeast, so the spontaneous generation theory was dismissed once and for all, though this did not answer how the first life form came on earth

Oparin, Haldane and Miller

- Oparin of Russia and Haldane of England proposed that the first form of life could have come from pre-existing non-living organic molecules such as RNA and protein, and that the formation of life was preceded by chemical evolution, that is the formation of diverse organic molecules from inorganic constituents
- the conditions on earth were high temperature, volcanic storms and a reducing atmosphere containing CH_4 and NH_3
- in 1953 S.L. Miller, an American scientist, created similar conditions on a laboratory scale, creating an electric discharge in a closed flask containing CH_4, H_2, NH_3 and water vapour at 800 degrees C, and observed the formation of amino acids
- in similar experiments others observed the formation of sugars, nitrogen bases, pigment and fats, and analysis of meteorite content also revealed similar compounds, indicating that similar processes are occurring elsewhere in space, so with this limited evidence the first part of the conjectured story, chemical evolution, was more or less accepted

A closed flask, holding methane, hydrogen, ammonia and water vapour

Electric discharge passed through the mixture

The whole held at 800 degrees C

Amino acids found in the condensate

Figure 6.A Diagrammatic representation of Miller's experiment

6.3 The first forms of life

From molecules to cells

- we have no idea about how the first self replicating metabolic capsule of life arose
- the first non-cellular forms of life could have originated 3 billion years back, and would have been giant molecules such as RNA, protein and polysaccharides, these capsules reproducing their molecules perhaps
- the first cellular form of life did not possibly originate till about 2000 million years ago, and these were probably single-cells, all life forms being in a water environment only
- this version of a biogenesis, that the first form of life arose slowly through evolutionary forces from non-living molecules, is accepted by the majority

6.4 Evolution of life forms, a theory

Special creation, and what challenged it

- conventional religious literature tells us about the theory of special creation, which has three connotations, that all living organisms we see today were created as such, that the diversity was always the same since creation and will be the same in future, and that earth is about 4000 years old
- all these ideas were strongly challenged during the nineteenth century
- based on observations made during a sea voyage in a sail ship called H.M.S. Beagle round the world, Charles Darwin concluded that existing living forms share similarities to varying degrees not only among themselves but also with life forms that existed millions of years ago, many of which do not exist any more
- there had been extinctions of different life forms in the years gone by, just as new forms of life arose at different periods of the history of earth, so there has been gradual evolution of life forms

6.5 Natural selection

Variation and fitness

- any population has built in variation in characteristics, and those characteristics which enable some to survive better in natural conditions of climate, food and physical factors would outbreed others that are less endowed to survive under such conditions
- another word used is the fitness of the individual or population, and fitness according to Darwin refers ultimately and only to reproductive fitness, so those who are better fit in an environment leave more progeny than others, will survive more, and hence are selected by nature, which he called natural selection and implied as a mechanism of evolution
- Alfred Wallace, a naturalist who worked in the Malay Archipelago, had also come to similar conclusions around the same time
- in due course of time apparently new types of organism are recognisable, all existing life forms sharing similarities and common ancestors, though these ancestors were present at different periods in the history of earth, that is epochs, periods and eras, and the geological history of earth closely correlates with its biological history, a common permissible conclusion being that earth is billions of years old

6.6 The evidences for evolution

Fossils

- fossils are remains of hard parts of life-forms found in rocks, rocks form sediments, and a cross-section of earth's crust indicates the arrangement of sediments one over the other during the long history of earth
- different-aged rock sediments contain fossils of different life-forms which probably died during the formation of that particular sediment, some of them appearing similar to modern organisms and some representing extinct organisms such as the dinosaurs
- a study of fossils in different sedimentary layers indicates the geological period in which they existed, and showed that life-forms varied over time and that certain life forms are restricted to certain geological time-spans, so new forms of life have arisen at different times in the history of earth, all of which is called paleontological evidence

Embryology, anatomy and morphology

- embryological support for evolution was also proposed by Ernst Heckel, based on the observation of certain features during the embryonic stage common to all vertebrates that are absent in the adult, for example the embryos of all vertebrates including human develop a row of vestigial gill slits just behind the head, but these are functional only in fish and not found in any other adult vertebrates
- this proposal was disapproved on careful study performed by Karl Ernst von Baer, who noted that embryos never pass through the adult stages of other animals
- comparative anatomy and morphology shows similarities and differences among organisms of today and those that existed years ago, and such similarities can be interpreted to understand whether common ancestors were shared or not

A family tree of the dinosaurs

Their living modern day counterparts alongside

Crocodiles on one branch

Birds on another

Figure 6.B A family tree of dinosaurs and their living modern day counterpart organisms like crocodiles and birds

6.7 Homology

The same bones, different work

- whales, bats, cheetah and human, all mammals, share similarities in the pattern of bones of the forelimbs
- though these forelimbs perform different functions in these animals they have a similar anatomical structure, all of them having humerus, radius, ulna, carpals, metacarpals and phalanges
- in these animals the same structure developed along different directions due to adaptations to different needs, which is divergent evolution, and these structures are homologous
- homology indicates common ancestry, other examples being vertebrate hearts or brains

Thorn and tendril of Bougainvillea and Cucurbita, in plants

Forelimbs of whale, bat, cheetah and human, in animals

The same set of bones in each

Developed along different directions for different needs

Figure 6.C Example of homologous organs in (a) Plants and (b) Animals

6.8 Analogy

Convergent evolution

- in plants also, the thorn and tendrils of Bougainvillea and Cucurbita represent homology, and homology is based on divergent evolution whereas analogy refers to a situation exactly opposite
- wings of butterfly and of birds look alike but are not anatomically similar structures, though they perform similar functions, so analogous structures are a result of convergent evolution, different structures evolving for the same function and hence having similarity
- other examples of analogy are the eye of the octopus and of mammals, or the flippers of penguins and dolphins, and one can say that it is the similar habitat that has resulted in selection of similar adaptive features in different groups of organisms but toward the same function, sweet potato as a root modification and potato as a stem modification being another example
- in the same line of argument, similarities in proteins and genes performing a given function among diverse organisms give clues to common ancestry, these biochemical similarities pointing to the same shared ancestry as structural similarities

6.9 What breeding shows

Man as a selector

- man has bred selected plants and animals for agriculture, horticulture, sport or security, and has domesticated many wild animals and crops
- this intensive breeding programme has created breeds that differ from other breeds, as in dogs, but are still of the same group
- it is argued that if within hundreds of years man could create new breeds, nature could have done the same over millions of years

The moths of England

- in a collection of moths made in the 1850s, that is before industrialisation set in, it was observed that there were more white-winged moths on trees than dark-winged or melanised moths
- however, in a collection carried out from the same area but after industrialisation, that is in 1920, there were more dark-winged moths in the same area
- the proportion was reversed

White-winged moth and dark-winged melanised moth

On a tree trunk in an unpolluted area, pale with lichen

On a tree trunk in a polluted area, darkened by soot

Each moth conspicuous against the wrong background

Figure 6.D Figure showing white - winged moth and dark - winged moth (melanised) on a tree trunk (a) In unpolluted area (b) In polluted area

6.10 Why the moths changed

The predator explanation

- the explanation put forth was that predators will spot a moth against a contrasting background
- during the post-industrialisation period the tree trunks became dark due to industrial smoke and soot, so under this condition the white-winged moth did not survive due to predators, while the dark-winged or melanised moth survived
- before industrialisation set in, a thick growth of almost white-coloured lichen covered the trees, and in that background the white winged moth survived but the dark-coloured moths were picked out by predators, lichens being usable as industrial pollution indicators since they will not grow in areas that are polluted
- so moths that were able to camouflage themselves survived, which is supported by the fact that in areas where industrialisation did not occur, such as rural areas, the count of melanic moths was low, showing that in a mixed population those that can better adapt survive and increase in population size, though no variant is completely wiped out

Evolution by our own doing

- excess use of herbicides and pesticides has only resulted in selection of resistant varieties in a much shorter time scale
- this is also true for microbes against which we employ antibiotics, or drugs against eukaryotic organisms and cells, so resistant organisms and cells are appearing in a time scale of months or years and not centuries
- these are examples of evolution by anthropogenic action, and this also tells us that evolution is not a directed process in the sense of determinism, but a stochastic process based on chance events in nature and chance mutation in the organisms

6.11 Adaptive radiation

The Galapagos finches

- during his journey Darwin went to the Galapagos Islands, where he observed an amazing diversity of creatures
- of particular interest, small black birds later called Darwin's finches amazed him
- he realised that there were many varieties of finches in the same island

Beaks of the finches Darwin found

The original seed-eating form

Insectivorous forms with altered beaks

Vegetarian forms with altered beaks

Figure 6.E Variety of beaks of finches that Darwin found in Galapagos Island

A number of marsupials, each different from the other

All evolved from an ancestral stock

All within the Australian island continent

Figure 6.F Adaptive radiation of marsupials of Australia

6.12 What adaptive radiation is

Radiating from a point

- all the varieties, Darwin conjectured, evolved on the island itself, and from the original seed-eating features many other forms with altered beaks arose, enabling them to become insectivorous and vegetarian finches
- this process of evolution of different species in a given geographical area, starting from a point and literally radiating to other areas of geography or habitats, is called adaptive radiation, and Darwin's finches represent one of the best examples of this phenomenon
- another example is the Australian marsupials, a number of them, each different from the other, having evolved from an ancestral stock but all within the Australian island continent
- when more than one adaptive radiation appears to have occurred in an isolated geographical area, representing different habitats, one can call this convergent evolution

Australian marsupials on one side

Placental mammals on the other

Each marsupial matched by a similar placental form

Placental wolf beside Tasmanian wolf, the marsupial

Figure 6.G Picture showing convergent evolution of Australian Marsupials and placental mammals

6.13 Biological evolution

Where it began

- placental mammals in Australia also exhibit adaptive radiation, evolving into varieties of placental mammals each of which appears similar to a corresponding marsupial, such as the placental wolf and the Tasmanian wolf
- evolution by natural selection in a true sense would have started when cellular forms of life with differences in metabolic capability originated on earth
- the essence of Darwinian theory about evolution is natural selection, and the rate of appearance of new forms is linked to the life cycle or the life span

Why microbes evolve fast

- microbes that divide fast have the ability to multiply and become millions of individuals within hours, and a colony of bacteria growing on a given medium has built-in variation in terms of ability to utilise a feed component
- a change in the medium composition would bring out only that part of the population that can survive under the new conditions, and in due course of time this variant population outgrows the others and appears as a new species, which would happen within days
- for the same thing to happen in a fish or fowl would take millions of years, as the life spans of these animals are in years, and here we say that the fitness of the variant is better than that of the original under the new conditions
- nature selects for fitness, and the so-called fitness is based on characteristics which are inherited, so there must be a genetic basis for getting selected and to evolve, another way of saying which is that some organisms are better adapted to survive in an otherwise hostile environment, adaptive ability being inherited and having a genetic basis, so fitness is the end result of the ability to adapt and get selected by nature

6.14 Branching descent, and Lamarck

The two key concepts

- branching descent and natural selection are the two key concepts of the Darwinian theory of evolution
- even before Darwin, a French naturalist Lamarck had said that evolution of life forms had occurred, but driven by use and disuse of organs
- he gave the example of giraffes, who in an attempt to forage leaves on tall trees had to adapt by elongation of their necks, and as they passed on this acquired character of elongated neck to succeeding generations, giraffes slowly over the years came to acquire long necks, though nobody believes this conjecture any more

Process or result

- the world we see, inanimate and animate, is only the success stories of evolution
- when we describe the story of this world we describe evolution as a process, whereas when we describe the story of life on earth we treat evolution as a consequence of a process called natural selection
- we are still not very clear whether to regard evolution and natural selection as processes or as end results of unknown processes

Malthus, and Darwin's insight

- it is possible that the work of Thomas Malthus on populations influenced Darwin, and natural selection is based on certain observations which are factual, that natural resources are limited, that populations are stable in size except for seasonal fluctuation, that members of a population vary in characteristics even though they look superficially similar, and that most variations are inherited
- the fact that theoretically population size will grow exponentially if everybody reproduced maximally, and the fact that population sizes in reality are limited, means that there had been competition for resources, and only some survived and grew at the cost of others that could not flourish
- the novelty and brilliant insight of Darwin was this, that he asserted that variations which are heritable and which make resource utilisation better for a few, better adapted to the habitat, will enable only those to reproduce and leave more progeny, so over many generations survivors will leave more progeny and there would be a change in population characteristic, and hence new forms appear to arise

6.15 Mechanism of evolution

deVries and mutation

- even though Mendel had talked of inheritable factors influencing phenotype, Darwin either ignored these observations or kept silent
- in the first decade of the twentieth century Hugo deVries, based on his work on evening primrose, brought forth the idea of mutations, large differences arising suddenly in a population, and believed that it is mutation which causes evolution and not the minor heritable variations that Darwin talked about
- mutations are random and directionless while Darwinian variations are small and directional, and evolution for Darwin was gradual while deVries believed mutation caused speciation and hence called it saltation, a single step large mutation, studies in population genetics later bringing out some clarity

Stabilising selection, more individuals at the mean value

Directional selection, more individuals away from the mean

Disruptive selection, more individuals at both extremes

The distribution curve reshaped in each case

Figure 6.H Diagrammatic representation of the operation of natural selection on different traits : (a) Stabilising (b) Directional and (c) Disruptive

6.16 Hardy-Weinberg principle

Genetic equilibrium

- in a given population one can find out the frequency of occurrence of alleles of a gene or a locus, and this frequency is supposed to remain fixed and even remain the same through generations, the Hardy-Weinberg principle stating it using algebraic equations
- the principle says that allele frequencies in a population are stable and constant from generation to generation
- the gene pool, that is the total genes and their alleles in a population, remains a constant, which is called genetic equilibrium
- the sum total of all the allelic frequencies is 1

6.17 The algebra, and the five factors

The binomial expansion

- individual frequencies can be named p, q and so on, and in a diploid p and q represent the frequency of allele A and allele a
- the frequency of AA individuals in a population is simply p^2, which stated another way is that the probability that an allele A with a frequency of p appears on both the chromosomes of a diploid individual is simply the product of the probabilities
- similarly the frequency of aa is q^2 and of Aa is 2pq, hence $p^2 + 2pq + q^2 = 1$, which is a binomial expansion of $(p+q)^2$
- when a frequency measured differs from the expected value, the difference, and its direction, indicates the extent of evolutionary change, so a disturbance in genetic equilibrium, that is a change of frequency of alleles in a population, would be interpreted as resulting in evolution

What disturbs the equilibrium

- five factors are known to affect Hardy-Weinberg equilibrium, gene migration or gene flow, genetic drift, mutation, genetic recombination and natural selection
- when migration of a section of a population to another place and population occurs, gene frequencies change in the original as well as in the new population, new genes and alleles being added to the new population and lost from the old one, and there would be a gene flow if this gene migration happens multiple times
- if the same change occurs by chance it is called genetic drift, and sometimes the change in allele frequency is so different in the new sample of population that they become a different species, the original drifted population becoming founders and the effect being called the founder effect

Selection and speciation

- microbial experiments show that pre-existing advantageous mutations, when selected, will result in the observation of new phenotypes, and over a few generations this would result in speciation
- natural selection is a process in which heritable variations enabling better survival are enabled to reproduce and leave a greater number of progeny, and a critical analysis makes us believe that variation due to mutation, or due to recombination during gametogenesis, or due to gene flow or genetic drift, results in changed frequency of genes and alleles in future generations
- coupled to enhanced reproductive success, natural selection makes it look like a different population, and it can lead to stabilisation, in which more individuals acquire the mean character value, to directional change, in which more individuals acquire a value other than the mean, or to disruption, in which more individuals acquire peripheral character values at both ends of the distribution curve

6.18 A brief account of evolution

The first cells

- about 2000 million years ago the first cellular forms of life appeared on earth
- the mechanism of how non-cellular aggregates of giant macromolecules could evolve into cells with a membranous envelope is not known
- some of these cells had the ability to release O_2

Plant forms set against the geological periods

Algae and the earliest land plants at the base

Giant ferns and the pteridophytes above them

Seed plants and the flowering plants latest of all

Figure 6.1 A sketch of the evolution of plant forms through geological periods

6.19 Out of the water

The first invaders of land

- the reaction could have been similar to the light reaction in photosynthesis, where water is split with the help of solar energy captured and channelised by appropriate light harvesting pigments
- slowly single-celled organisms became multi-cellular life forms, and by the time of 500 million years ago invertebrates were formed and active, while jawless fish probably evolved around 350 million years ago and sea weeds and a few plants existed probably around 320 million years ago
- we are told that the first organisms that invaded land were plants, and they were widespread on land when animals invaded land
- fish with stout and strong fins could move on land and go back to water, about 350 million years ago, and in 1938 a fish caught in South Africa happened to be a Coelacanth, which was thought to be extinct, these animals called lobefins having evolved into the first amphibians

The vertebrates set against the geological periods

Jawless fish, then the bony fish

Amphibians, then the reptiles

Birds and mammals latest of all

Figure 6.J Representative evolutionary history of vertebrates through geological periods

6.20 From amphibian to reptile

The line of descent

- the first amphibians lived on both land and water, and there are no specimens of these left with us, however these were ancestors of modern day frogs and salamanders
- the amphibians evolved into reptiles, which lay thick-shelled eggs that do not dry up in the sun, unlike those of amphibians
- again we only see their modern day descendents, the turtles, tortoises and crocodiles

6.21 The age of reptiles

Dinosaurs

- in the next 200 million years or so, reptiles of different shapes and sizes dominated on earth, and giant ferns, the pteridophytes, were present but all fell to form coal deposits slowly
- some of these land reptiles went back into water to evolve into fish-like reptiles probably 200 million years ago, such as the Ichthyosaurs, while the land reptiles were the dinosaurs, the biggest of them, Tyrannosaurus rex, being about 20 feet in height and having huge fearsome dagger-like teeth
- about 65 million years ago the dinosaurs suddenly disappeared from the earth, and we do not know the true reason, some saying climatic changes killed them and some that most of them evolved into birds, while the truth may lie in between, and small sized reptiles of that era still exist today

The mammals

- the first mammals were like shrews, and their fossils are small sized, while mammals were viviparous and protected their unborn young inside the mother's body, and were more intelligent in sensing and avoiding danger at least
- when reptiles declined mammals took over this earth, and there were in South America mammals resembling horse, hippopotamus, bear and rabbit, but due to continental drift, when South America joined North America, these animals were overridden by North American fauna
- due to the same continental drift the pouched mammals of Australia survived, because of lack of competition from any other mammal
- some mammals live wholly in water, whales, dolphins, seals and sea cows being some examples, while the evolution of horse, elephant and dog are special stories of evolution, and the most successful story is the evolution of man with language skills and self-consciousness

6.22 Origin and evolution of man

The first hominids

- about 15 million years ago primates called Dryopithecus and Ramapithecus were existing, hairy and walking like gorillas and chimpanzees, Ramapithecus being more man-like while Dryopithecus was more ape-like
- a few fossils of man-like bones have been discovered in Ethiopia and Tanzania, which revealed hominid features leading to the belief that about 3 to 4 million years ago man-like primates walked in eastern Africa, probably not taller than 4 feet but walking upright
- two million years ago Australopithecines probably lived in East African grasslands, and evidence shows they hunted with stone weapons but essentially ate fruit
- some of the bones among those discovered were different, and this creature, called the first human-like being, the hominid, was called Homo habilis, with brain capacities between 650-800cc, and probably did not eat meat

Skull of an adult modern human being

Skull of a baby chimpanzee

Skull of an adult chimpanzee

The baby chimpanzee skull closer to the adult human than to the adult chimpanzee

Figure 6.K A comparison of the skulls of adult modern human being, baby chimpanzee and adult chimpanzee. The skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull

6.23 The later humans

Erectus, Neanderthal, sapiens

- fossils discovered in Java in 1891 revealed the next stage, Homo erectus, about 1.5 million years ago, with a large brain around 900cc, and Homo erectus probably ate meat
- the Neanderthal man, with a brain size of 1400cc, lived in the near east and central Asia between 1,00,000 and 40,000 years back, used hides to protect their body and buried their dead
- Homo sapiens arose in Africa and moved across continents and developed into distinct races, and during the ice age between 75,000 and 10,000 years ago modern Homo sapiens arose
- pre-historic cave art developed about 18,000 years ago, one such cave painting by pre-historic humans being seen at the Bhimbetka rock shelter in Raisen district of Madhya Pradesh, while agriculture came around 10,000 years back and human settlements started, the rest being part of human history of growth and decline of civilisations

6.24 Summary

What the chapter has covered

- the origin of life on earth can be understood only against the background of the origin of the universe, especially the earth, and most scientists believe chemical evolution, that is the formation of biomolecules, preceded the appearance of the first cellular forms of life
- the subsequent events as to what happened to the first form of life are a conjectured story based on Darwinian ideas of organic evolution by natural selection, and the diversity of life forms on earth has been changing over millions of years
- it is generally believed that variations in a population result in variable fitness, and other phenomena such as habitat fragmentation and genetic drift may accentuate these variations, leading to the appearance of new species and hence evolution
- homology is accounted for by the idea of branching descent, the study of comparative anatomy, fossils and comparative biochemistry provides evidence for evolution, and among the stories of evolution of individual species the story of the evolution of modern man is most interesting and appears to parallel the evolution of the human brain and language

6.25 Exercises

From this chapter

1. Explain antibiotic resistance observed in bacteria in light of Darwinian selection theory.
2. Find out from newspapers and popular science articles any new fossil discoveries or controversies about evolution.
3. Attempt giving a clear definition of the term species.
4. Try to trace the various components of human evolution (hint: brain size and function, skeletal structure, dietary preference, etc.)
5. Find out through internet and popular science articles whether animals other than man has self-consciousness.
6. List 10 modern-day animals and using the internet resources link it to a corresponding ancient fossil. Name both.
7. Practise drawing various animals and plants.
8. Describe one example of adaptive radiation.
9. Can we call human evolution as adaptive radiation?
10. Using various resources such as your school Library or the internet and discussions with your teacher, trace the evolutionary stages of any one animal, say horse.